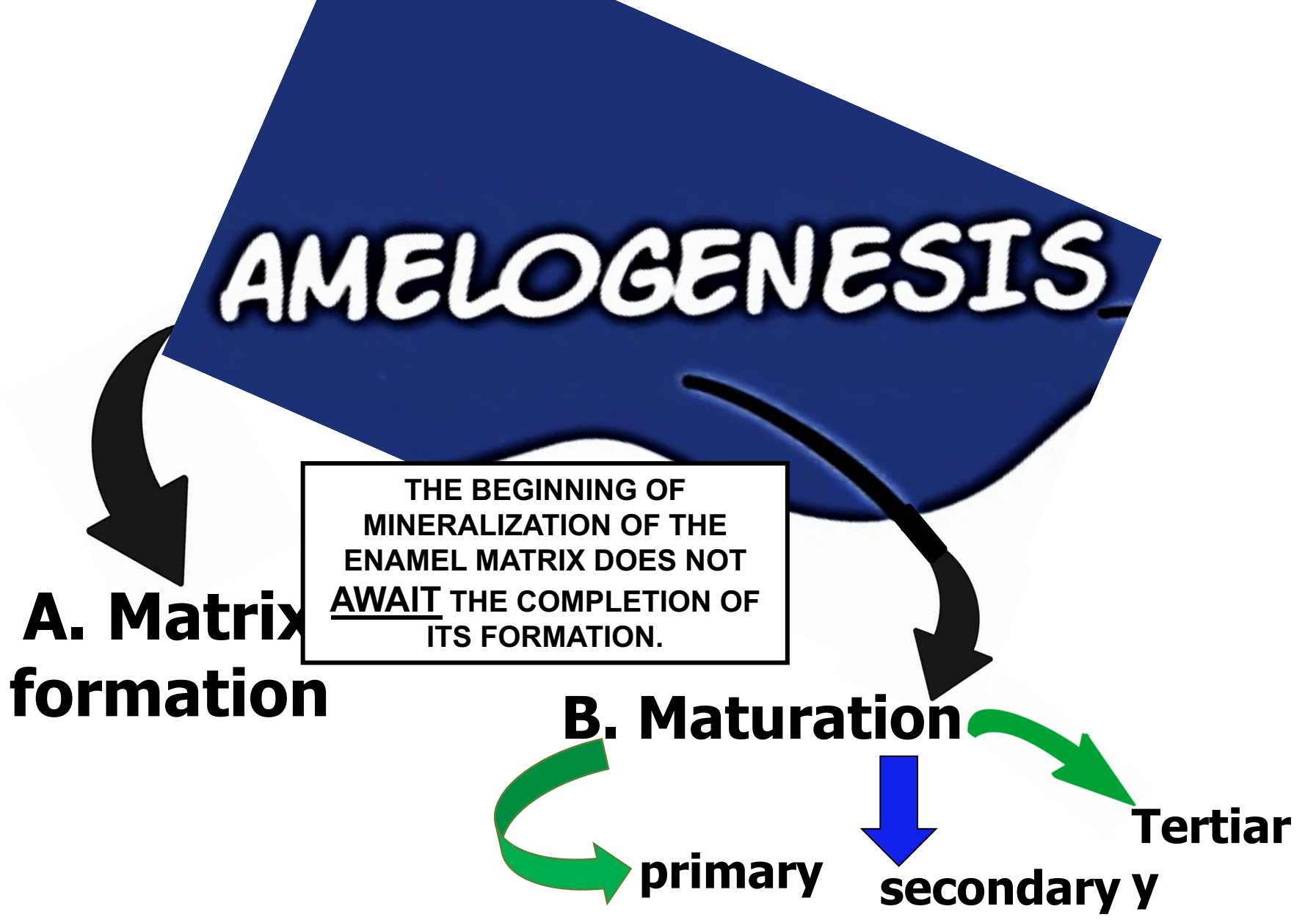


AMELOGENESIS



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graph TD; A[AMELOGENESIS] --> B[B. Maturation]; B --> C[primary]; B --> D[secondary]; B --> E[Tertiar];
```

The diagram illustrates the process of amelogenesis. At the top, a blue banner contains the word 'AMELOGENESIS'. A large black arrow points from this banner to the text 'A. Matrix formation'. Another large black arrow points from the banner to the text 'B. Maturation'. Below 'B. Maturation', three arrows branch out: a green curved arrow pointing to 'primary', a blue straight arrow pointing to 'secondary', and a green curved arrow pointing to 'Tertiar'. A central text box contains the statement: 'THE BEGINNING OF MINERALIZATION OF THE ENAMEL MATRIX DOES NOT AWAIT THE COMPLETION OF ITS FORMATION.'

THE BEGINNING OF
MINERALIZATION OF THE
ENAMEL MATRIX DOES NOT
AWAIT THE COMPLETION OF
ITS FORMATION.

**A. Matrix
formation**

B. Maturation

primary

secondary

Tertiar

Amelogenesis could be summarized as

1. Organic matrix formation.
2. Mineralization.

In amelogenesis there is no clear cut band of organic matrix such as predentin & osteoid tissues in dentin & bone.

Enamel crystallites appear almost *immediately* in newly secreted enamel matrix.

A.Matrix formation

1-Initial rodless (aprismatic) enamel layer

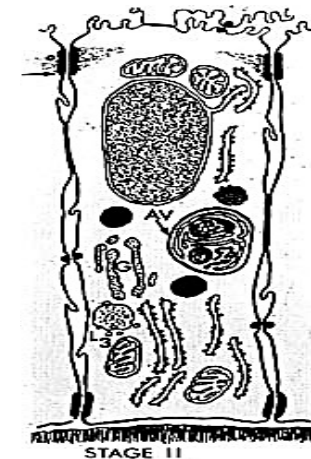
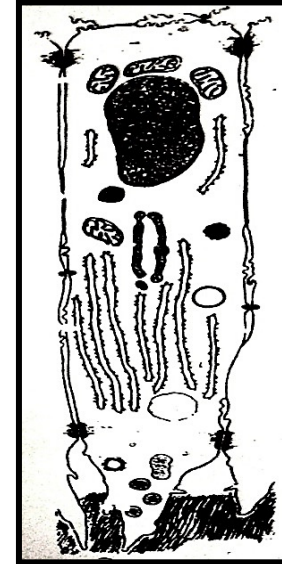
2-Rod (prismatic) enamel layer:

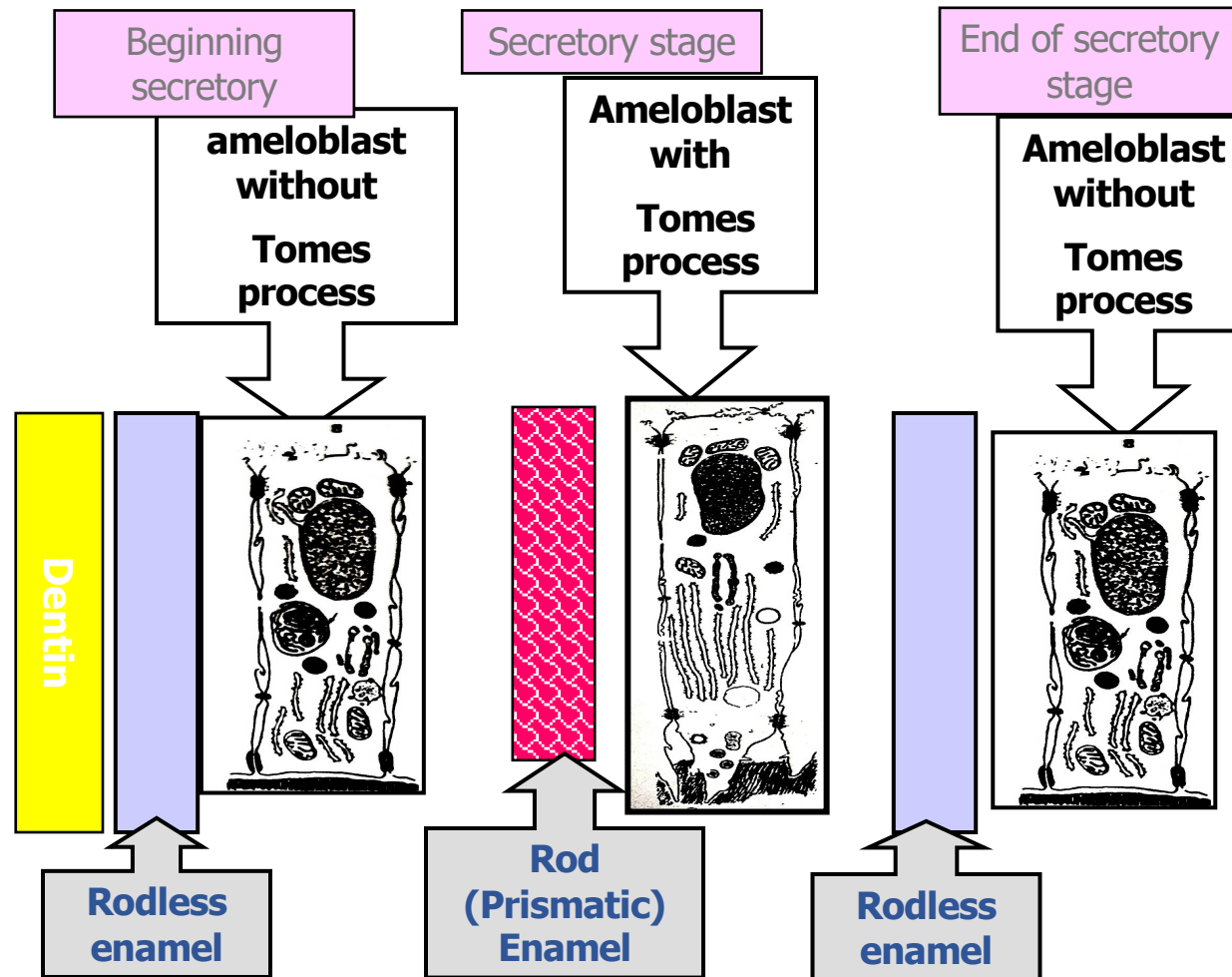
Ameloblasts become taller and has Tomes' processes

3-Final rodless (aprismatic) enamel layer

Ameloblasts become shorter and Tomes' processes are lost

End of secretory ameloblast





PRIMARY MATURATION

20-30% of mineral content

With matrix formation

SECONDARY MATURATION

- 1- Selective reabsorption of proteins
- Immature enamel : rich in **amelogenin**
- Mature Enamel : Rich in **Enamelin**

20-30%-----96%

Immature Enamel	Mature Enamel
25-30% In-organic	96% In-organic
Enamel proteins: Amelogenins and Ameloblastins	Enamel proteins: Enamelins and Tuftelins.

2- CRYSTAL GROWTH (CRESTALS NEVER FUSE)

Ameloblasts secrete mineral ions on the already formed crystals

leading to an increase in the thickness and width of crystals

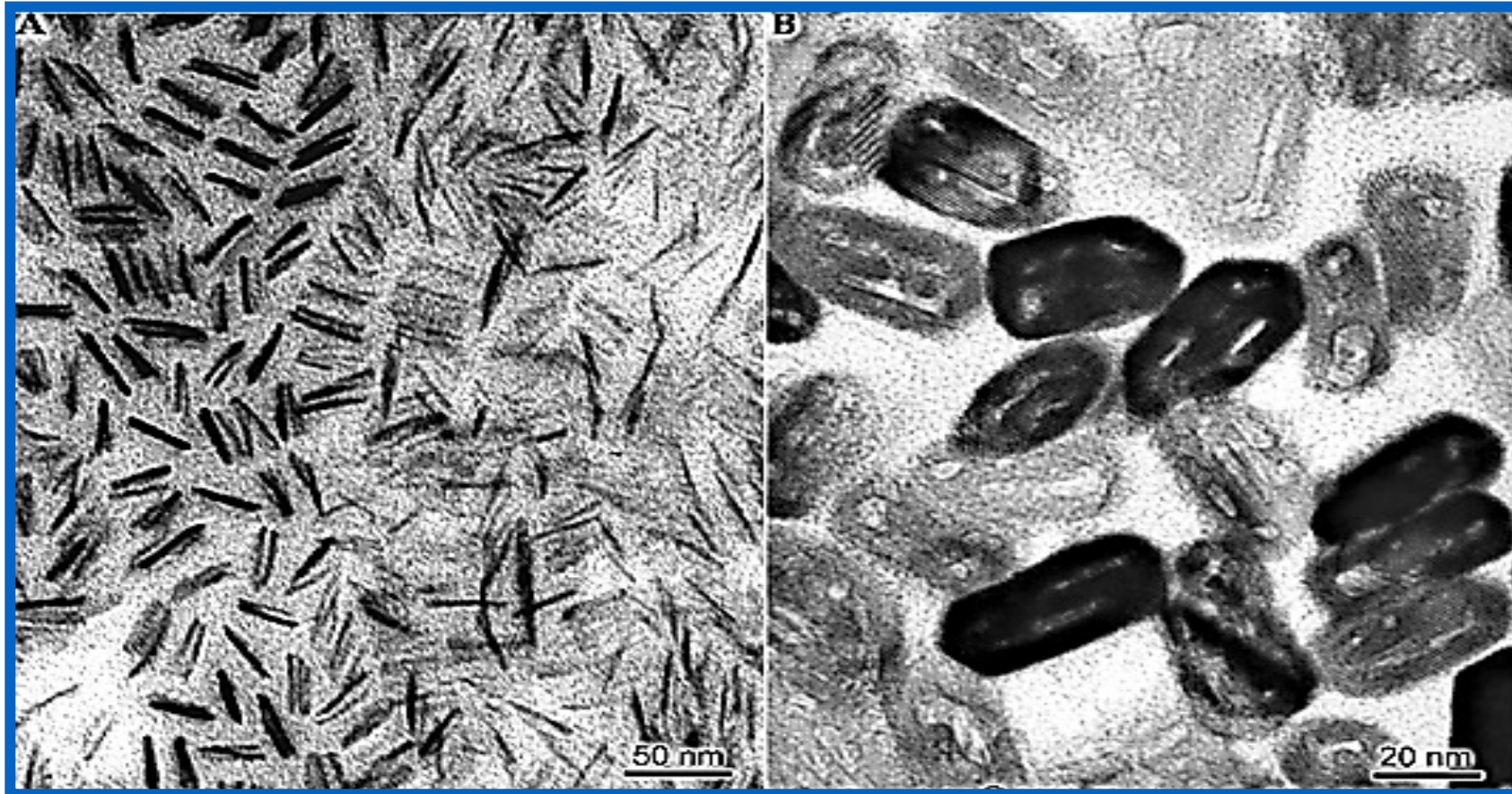
TERTIARY MATURATION

After eruption

Ions from saliva. 96-----98%

Mineralization

- a. Partial mineralization (25-30%).
- b. Maturation (gradual completion of mineralization up to 96%)



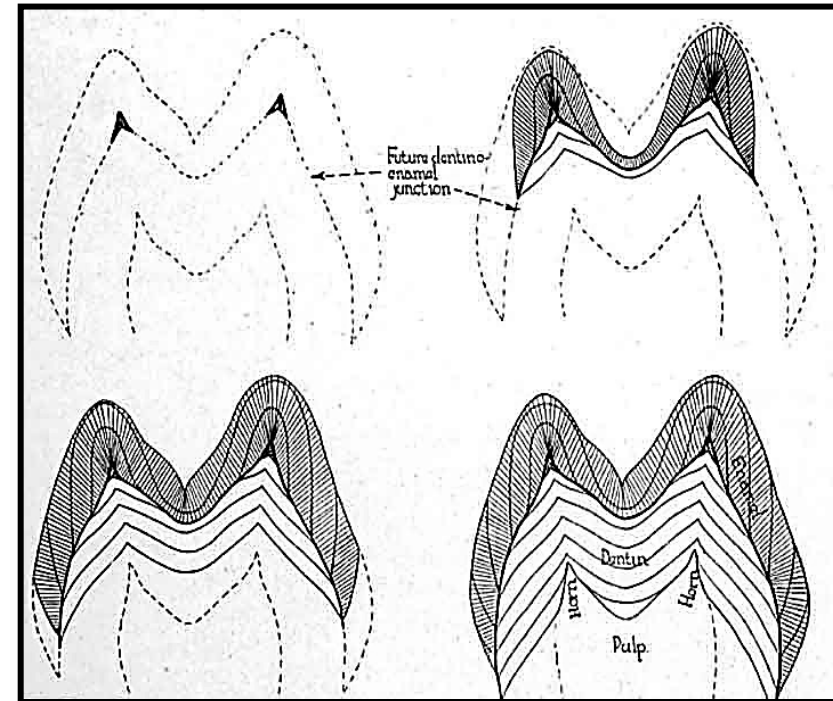
DIRECTIONS OF MATURATION

First Direction:

From cusp tip cervically

Second Direction:

From DEJ to the outer surface of enamel



- Interference during E. matrix formation may cause

E. hypoplasia.

- Interference during E. maturation may cause

E. hypocalcification.

- Each condition may be caused by systemic, local, or hereditary factors.

Age changes of Enamel

1- Attrition.

2-Permeability.

3- Color.

4- Modification in the surface layer.

1- Attrition

The most apparent age change in enamel.

A physiological wear of the tooth hard substances in (occlusal or incisal surfaces & proximal contact points) as a result of mastication physiologic tooth movement, functional or para functional movements of mandible.

(tooth to tooth contact)

The amount of wear varies with diet,
gender
hardness of teeth,
certain habits.
nature of occlusion



Clinical feature:-

1-Small **shiny** facets on cusp tips/ridges/ flattening of incisal ridges.

2-Because of slight mobility of teeth in their sockets & resiliency of PDL, facets also occur at proximal surfaces.



- 4-Decreased cusp height.
- 5-Flattening of occlusal plane.
- 6-Shortening of height of dental arch.



- All these changes occur more severely in men than women due to greater masticatory force.

permanent teeth more than deciduous teeth.

The nature of diet & some habits (**Bruxism**) may affect the degree of attrition



Abfraction

Recently, it has been proposed that the predominant causative factor of some of the cervical, wedge-shaped defects is a strong (heavy) eccentric occlusal force resulting in microfracture occur at the cervical area of the tooth.

This defects is termed
as **idiopathic erosion**
or **abfraction.**



Erosion

The loss of hard dental tissue *chemically* e.g. lemon juice, beverage.

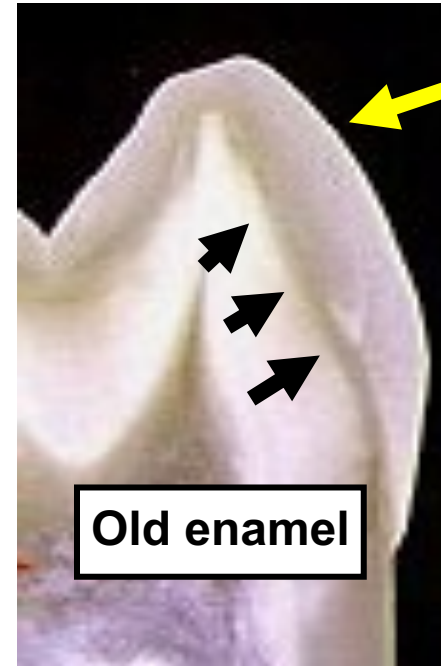
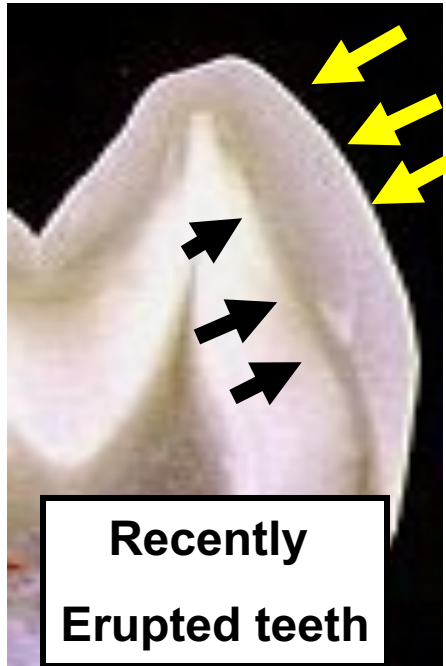


2-Permeability

Enamel is semi-permeable, allowing ionic exchange which leads to its **surface modification**.

Enamel crystals may **acquire more ions** reducing the size of the pores between them. This leads to decreased **permeability**.

The permeability **decreases** or even **disappears** from the outer surface, while that from the dentinal surface may remain.



3- Color changes of the tooth

Normally with age it become more yellowish as minerals more percipitated & also may it associated with internal calcification.

It may also be due to attrition exposing the yellow color of dentin(which become thicker & sclerotic) through the thinning layer of enamel with reducing its translucency.

Teeth color gradually darken with age due to the addition of organic material to enamel from the environment (tobacco, coffee & tea) Due to permeability.



4- Modification in the surface layer

After tooth eruption, enamel undergo modification in the surface layer (ionic exchange) between enamel & saliva (alteration in the chemical composition of enamel). This may lead to localized increase of certain elements (e.g. fluorine & nitrogen) in the superficial enamel layer by age.

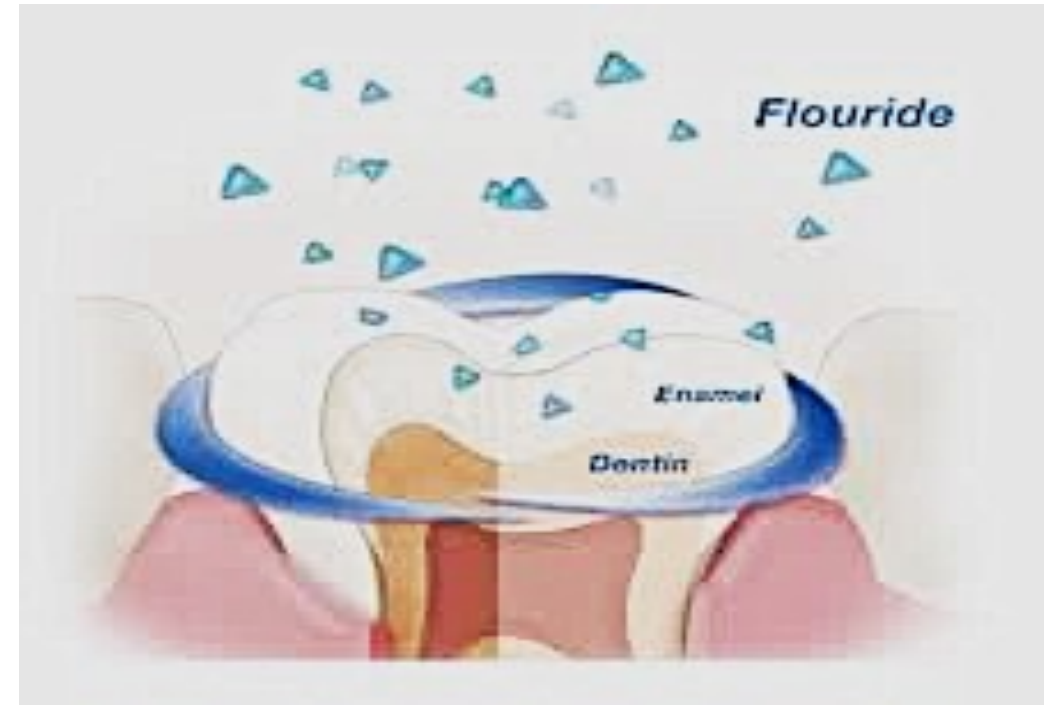
This exchange increases the inorganic content to be 98% called (2ry maturation)

A decrease of the **water content** of enamel by age was also suggested.

These alternations may lead to

reduction in the enamel porosity & caries incidence.

Which means that enamel near the surface become **decay resistant**.



Histologically,

- **The perikymata lines are lost.**

CLINICAL CONSIDERATION



1-Fluoridation:

fluoride ion is incorporated into crystal, they become more resistant to acid dissolution (caries) especially **the white spots**.

Also the presence of fluoride enhances precipitation of calcium phosphates.



This can be during E formation or even after eruption by Topical application.

The amount of fluoride must be controlled because high fluoride can cause mottled enamel (in excess of 5ppm)



Dental Fluorosis (mottled enamel)

2-Acid etching:

very important technique for conditioning enamel for many clinical procedures.

Used when doing

fissure sealants,

bonding of restorative materials to E ,

cementing of orthodontics brackets to tooth.

It is carried out by using a mild acid like ortho-phosphoric acid on the enamel surface for a controlled period of time.



-Remove the plaque & debris with a thin layer of E .

Then increase the porosity of exposed surfaces by **selective dissolution of crystals** to give better bonding surface for the restorative& adhesive materials.